

PRAKTIKUM BASIS DATA

MODUL 13

NoSQL_Document_Database



Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

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JURNAL

1. Tambah Toko

Jawaban:

```
>_ mongosh: jm13.pey56vy.... +
>_MONGOSH
< switched to db.jurnal13
> db.toko.insertMany([
  {
    name: "Toko Berkah",
    category: "Elektronik",
    city: "Jakarta",
    address: {
      street: "Jl. Sudirman",
      zipcode: "10220"
    },
    products: [
      { item: "TV", price: 3500000, stock: 10 },
      { item: "Kulkas", price: 2800000, stock: 5 }
    ]
  },
  {
    name: "Toko Maju",
    category: "Elektronik",
    city: "Bandung",
    address: {
      street: "Jl. Asia Afrika",
      zipcode: "40111"
    },
    products: [
      { item: "HP", price: 1500000, stock: 20 },
```

2. Tampilkan semua toko di Jakarta

Jawaban:

```
>_MONGOSH

> db.toko.find({
  city: "Jakarta"
}).pretty()
< {
  _id: ObjectId('6a26a7c771ae45d9245ddc7f'),
  name: 'Toko Berkah',
  category: 'Elektronik',
  city: 'Jakarta',
  address: {
    street: 'Jl. Sudirman',
    zipcode: '10220'
  },
  products: [
    {
      item: 'TV',
      price: 3500000,
      stock: 10
    },
    {
      item: 'Kulkas',
      price: 2800000,
      stock: 5
    }
  ]
}
```

3. Tampilkan toko yang memiliki produk dengan harga > 3.000.000

```

>_MONGOSH
> db.toko.find({
  "products.price": { $gt: 3000000 }
})
< {
  _id: ObjectId('6a26a7c771ae45d9245ddc7f'),
  name: 'Toko Berkah',
  category: 'Elektronik',
  city: 'Jakarta',
  address: {
    street: 'Jl. Sudirman',
    zipcode: '10220'
  },
  products: [
    {
      item: 'TV',
      price: 3500000,
      stock: 10
    },
    {
      item: 'Kulkas',
      price: 2800000,
      stock: 5
    }
  ]
}

```

4. Ubah category dan tambahkan produk ke Toko

```

db.toko.updateOne(
  { name: "Toko Berkah" },
  { $set: { category: "Aksesoris" } }
)
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
db.toko.updateOne(
  { name: "Toko Maju" },
  {
    $push: {
      products: {
        item: "Printer Epson",
        price: 3200000,
        stock: 5
      }
    }
  }
}

```

5. Hitung jumlah toko per kota (terbanyak dulu)

```
> db.toko.aggregate([
  {
    $group: {
      _id: "$city",
      jumlah_toko: { $sum: 1 }
    }
  },
  {
    $sort: {
      jumlah_toko: -1
    }
  }
])
< {
  _id: 'Jakarta',
  jumlah_toko: 3
}
{
  _id: 'Bandung',
  jumlah_toko: 2
}
{
  _id: 'Surabaya',
  jumlah_toko: 1
}
```

6. Hitung rata-rata harga produk per kategori

```
➤_ mongosh: jml3.peyb6vy.... +
>_MONGOSH
> db.toko.aggregate([
  {
    $unwind: "$products"
  },
  {
    $group: {
      _id: "$category",
      rata_rata_harga: {
        $avg: "$products.price"
      }
    }
  },
  {
    $sort: {
      rata_rata_harga: -1
    }
  }
])
< {
  _id: 'Elektronik',
  rata_rata_harga: 5595454.545454546
}
{
  _id: 'Aksesoris',
```

7. Tambahkan 1 produk baru

```
> db.toko.updateOne(
  { name: "Toko Maju" },
  {
    $push: {
      products: {
        item: "Mouse Gaming",
        price: 750000,
        stock: 12
      }
    }
  }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

8. Tampilkan harga produk tertinggi per kota


```

>_MONGOSH

> db.toko.aggregate([
  {
    $unwind: "$products"
  },
  {
    $group: {
      _id: "$city",
      harga_tertinggi: {
        $max: "$products.price"
      }
    }
  }
])
< {
  _id: 'Jakarta',
  harga_tertinggi: 77000000
}
{
  _id: 'Surabaya',
  harga_tertinggi: 120000000
}
{
  _id: 'Bandung',
  harga_tertinggi: 80000000
}

```

9. Pipeline lengkap (Elektronik → unwind → total stock per city → sort)

>_MONGOSH

```
> db.toko.aggregate([
  {
    $match: {
      category: "Elektronik"
    }
  },
  {
    $unwind: "$products"
  },
  {
    $group: {
      _id: "$city",
      total_stock: {
        $sum: "$products.stock"
      }
    }
  },
  {
    $sort: {
      total_stock: -1
    }
  }
])
< {
```

```

{
  _id: 'Bandung',
  total_stock: 63
}
{
  _id: 'Jakarta',
  total_stock: 33
}
{
  _id: 'Surabaya',
  total_stock: 29
}

```

10. Hapus toko dengan category Fashion dan city Surabaya

```

> db.toko.deleteOne({
  category: "Fashion",
  city: "Surabaya"
})
< {
  acknowledged: true,
  deletedCount: 0
}

```